

A PROJECT TITLE
ON
Insurance Claim Fraud Detection Using XG Boost with Grid/Bayesian Hyperparameter Search

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY

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by

BHARGAV	2520090169
HANISH	2520090107
TARUN	2520090020

Under the Esteemed guidance of

Dr. Y. Lakshmi
Prasanna

Designation, CSIT

DEPT. OF CS & IT



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



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Name of Title: **Insurance Claim Fraud Detection Using XG Boost with Grid/Bayesian Hyperparameter Search**

1. Abstract:

Insurance fraud has become one of the major challenges faced by the insurance industry, leading to significant financial losses and increased operational costs. Detecting fraudulent claims manually is a time-consuming and error-prone process due to the large volume of claims submitted every day. Machine Learning (ML) provides an effective solution by automatically identifying suspicious claims based on historical data and predictive analysis. This project presents an Insurance Claim Fraud Detection system using the Extreme Gradient Boosting (XGBoost) algorithm combined with Grid Search and Bayesian Hyperparameter Optimization techniques to improve model performance.

The dataset used in this project contains various customer, vehicle, policy, and accident-related attributes that help distinguish genuine claims from fraudulent ones. Initially, the dataset undergoes preprocessing steps including handling missing values, encoding categorical variables, feature scaling where necessary, and removing irrelevant attributes. Exploratory Data Analysis (EDA) is then performed to understand data distribution and identify relationships among features.

XGBoost is selected because of its high accuracy, efficient handling of structured datasets, robustness against overfitting, and capability to manage missing values. To further improve prediction accuracy, two hyperparameter optimization techniques are employed. Grid Search systematically evaluates predefined parameter combinations, while Bayesian Optimization intelligently explores the search space by learning from previous evaluations, thereby reducing computational cost. The optimized model is evaluated using performance metrics such as Accuracy, Precision, Recall, F1-Score, ROC-AUC Score, and Confusion Matrix.

Experimental results demonstrate that Bayesian Optimization achieves comparable or better predictive performance than Grid Search while requiring fewer iterations and less computational time. The proposed system successfully identifies fraudulent insurance claims with high precision and recall, making it suitable for real-world deployment. By automating fraud detection, insurance companies can reduce financial losses, accelerate claim processing, improve customer satisfaction, and support investigators in identifying suspicious activities more efficiently. This project highlights the practical application of advanced Machine Learning techniques in solving real-world fraud detection problems and demonstrates the effectiveness of XGBoost with optimized hyperparameters in achieving reliable and scalable insurance fraud detection.

2. INTRODUCTION:

Insurance fraud is a major issue that causes significant financial losses for insurance companies and increases costs for genuine customers. Detecting fraudulent claims manually is difficult due to the large volume of claims and complex fraud patterns. Machine Learning provides an efficient solution by analyzing historical claim data and identifying suspicious claims automatically. In this project, the XGBoost algorithm is used to classify insurance claims as genuine or fraudulent because of its high accuracy and efficiency. To further improve the model's performance, Grid Search and Bayesian Hyperparameter Optimization are applied to identify the optimal hyperparameter values. The model is evaluated using metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC, making it an effective and reliable solution for insurance claim fraud detection.

Among various Machine Learning algorithms, **Extreme Gradient Boosting (XGBoost)** is widely used because of its high accuracy, speed, and ability to handle structured datasets effectively. In this project, XGBoost is employed to classify insurance claims as genuine or fraudulent. To enhance the model's performance, **Grid Search** and **Bayesian Hyperparameter Optimization** are used to determine the optimal hyperparameter values. The trained model is evaluated using metrics such as **Accuracy, Precision, Recall, F1-Score, and ROC-AUC**, providing a reliable and efficient system for detecting fraudulent insurance claims.

3. SOFTWARE REQUIREMENTS:

- Programming Language: Python 3.10 or above
- Development Environment (IDE): Jupyter Notebook, Google Colab, or Visual Studio Code (VS Code)
- Machine Learning Library: XGBoost
- Hyperparameter Optimization: GridSearchCV and Optuna (Bayesian Optimization)
- Python Libraries:
 - Pandas
 - NumPy
 - Scikit-learn
 - Matplotlib
 - Seaborn
 - XGBoost
 - Optuna
- Dataset: Insurance Claim Fraud Detection Dataset
- Evaluation Metrics: Accuracy, Precision, Recall, F1-Score, ROC-AUC Score, and Confusion Matrix


Signature of the Guide


Course Instructor